

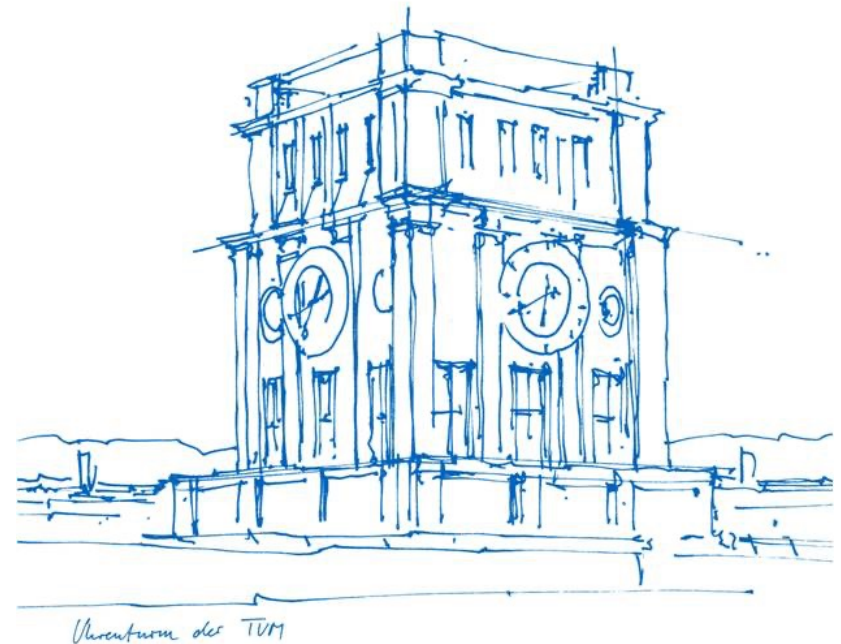
Probabilistic Forecasting of Regional Redispatch Volumes in the German Electricity Market

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Electricity Market Design & Congestion Management

Zonal model

- Large **bidding zones (BZ)** with a **uniform day-ahead price**.
- Considers only **aggregated zonal constraints**.
- Dispatch decisions are not physically feasible
→ **redispatch**: specific generators are instructed to produce more (**upward RD**) or less (**downward RD**) relative to their schedules
⇒ feasible market outcome

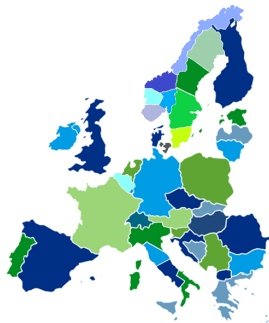


Illustration from: <https://fsr.eui.eu/electricity-markets-in-the-eu/>

Nodal model

- **Locational pricing**: one price per location
- Considers **local** supply and demand, as well as **transmission constraints**.
- Dispatch decisions are physically feasible
→ **no redispatch** is needed

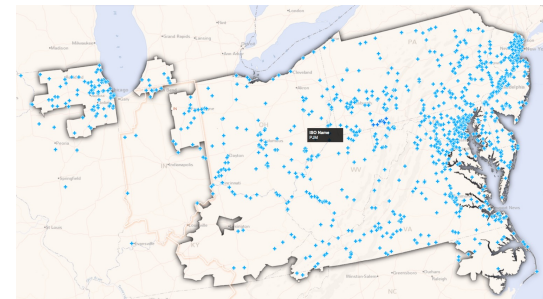


Illustration from: <https://www.energyacuity.com/blog/pjm-lmp-map/>

Germany: Critical BZ in the European Market

- Central position in the European grid
- Uneven geographical distribution of supply and demand
- Increasing integration of renewables
- Insufficient network capacity

➡ Congestion ➡ Redispatch

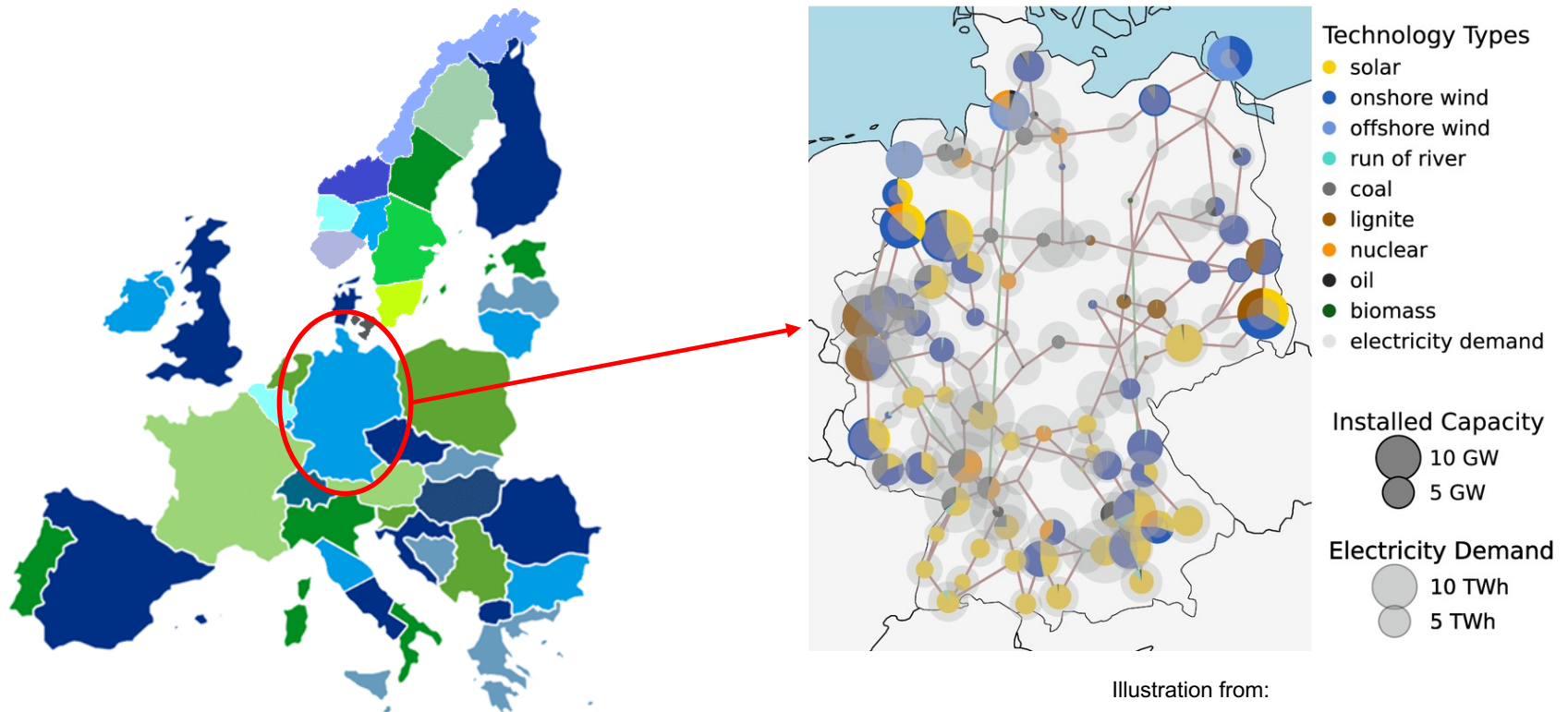
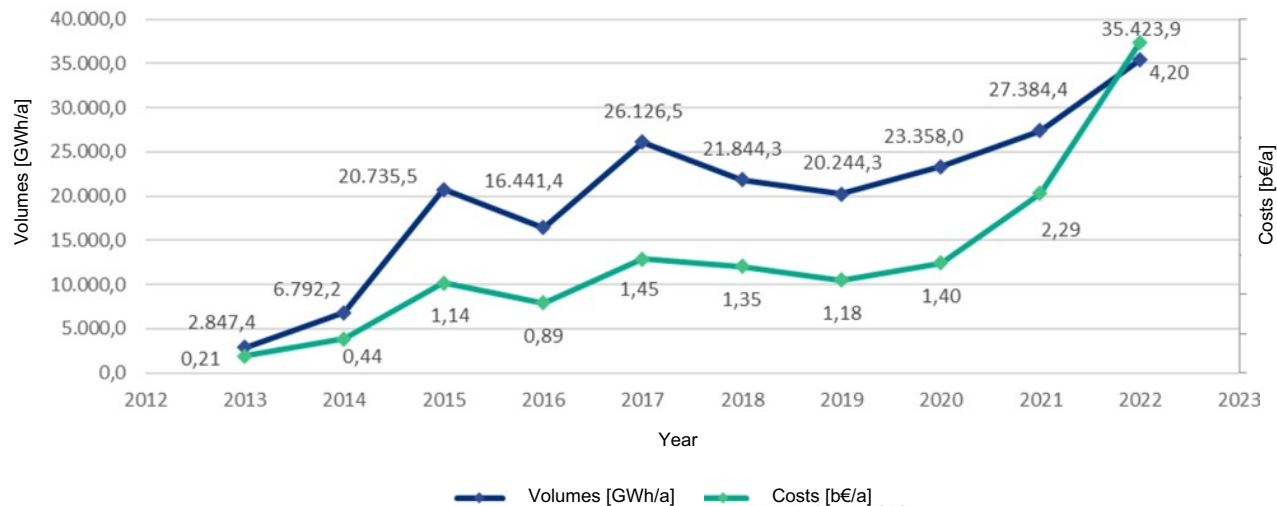


Illustration from:
*Net load forecasting using different
aggregation levels, 2022, Beichter et. al.*

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Redispatch 2.0: current methodology implemented by TSOs.

Illustration from: BDEW 2022, Bundesnetzagentur 2023c

From Redispatch 2.0 to Redispatch 3.0

Redispatch 2.0

- **Cost-based:** redispatched generators are paid based on their declared costs.
- Includes only generators with **capacity ≥ 100 kW**.
- Criticism:
 - Large RD costs & volumes
 - RES curtailment
 - No competition

Redispatch 3.0

- Aims for **automated, preventive, and market-based approaches**.
- Integrates smaller, decentralized units (e.g., EV, battery storage) – not only ≥ 100 kW!
- **Market-based integration of flexibility:**
 - Local flexibility markets: GOPACS, enera.
- Methodology is in progress.

Challenges of Redispatch 3.0

The integration of **market-based components** into **Redispatch 3.0** poses challenges:

→ Market participants need to develop **new bidding strategies** based on **expected redispatch**

Probabilistic forecasting of regional redispatch volumes



Data Acquisition & Geolocation

Obtain time series of RD volumes and unit locations.



Region Identification via Clustering

Cluster units by average RD volume and location.



Model Development

1. ARIMAX model with bootstrapped intervals
2. Transformer architecture with quantile regression

Data

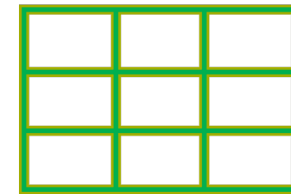


Netztransparenz.de



Open Power System Data

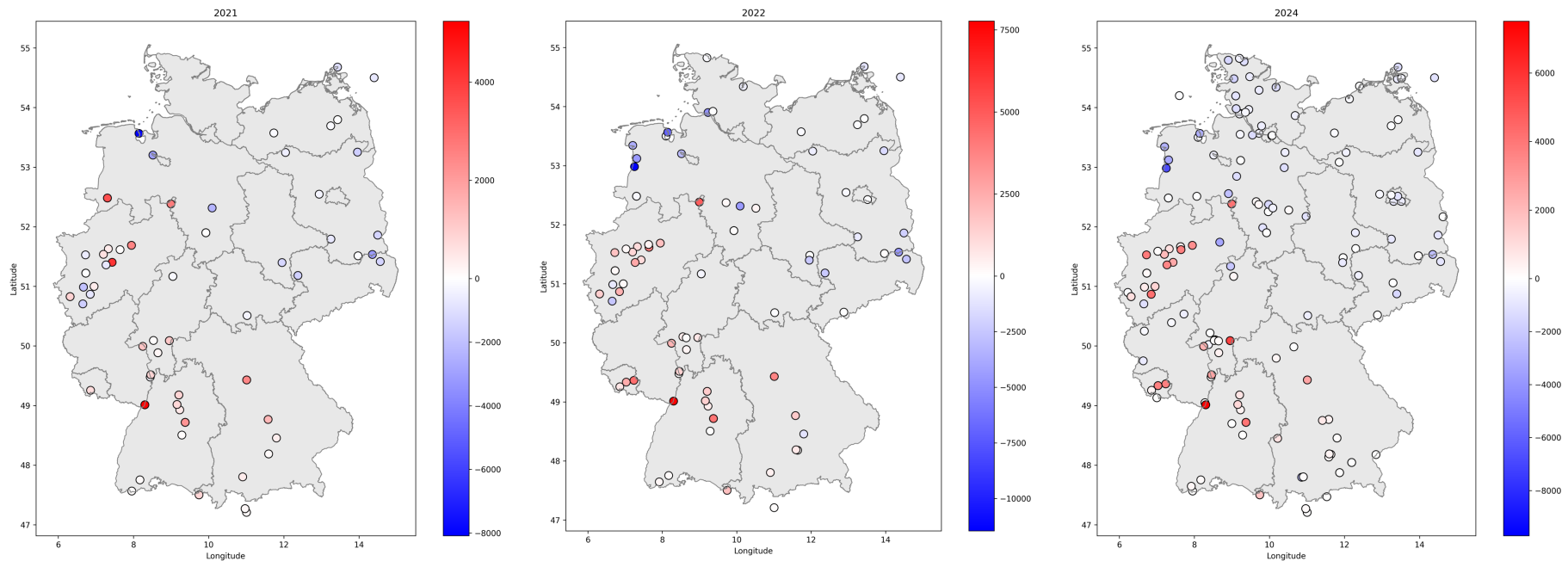
Fuzzy matching



Time series of **daily**
redispatch volumes per unit &
geographic coordinates

- Convention:
 - *Upward redispatch*: **positive** quantity
 - *Downward redispatch*: **negative** quantity
- Daily load, generation & day-ahead prices for Germany: ENTSO-E Transparency Platform.

Average Yearly Redispatch Volumes in 2021-2024



- Increasing trend: number of RD units, RD volumes
- Persistent spatial patterns:
 - North: downward RD
 - South: upward RD

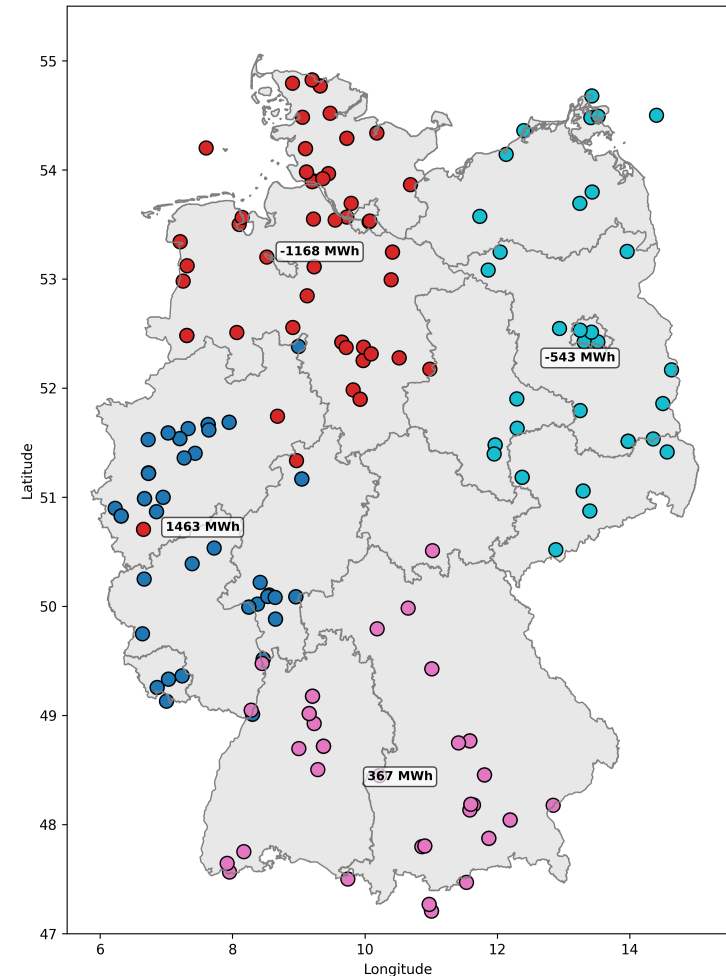
*Note: different scale in each year!
Values are in MWh.

Spatiotemporal Redispatch Patterns

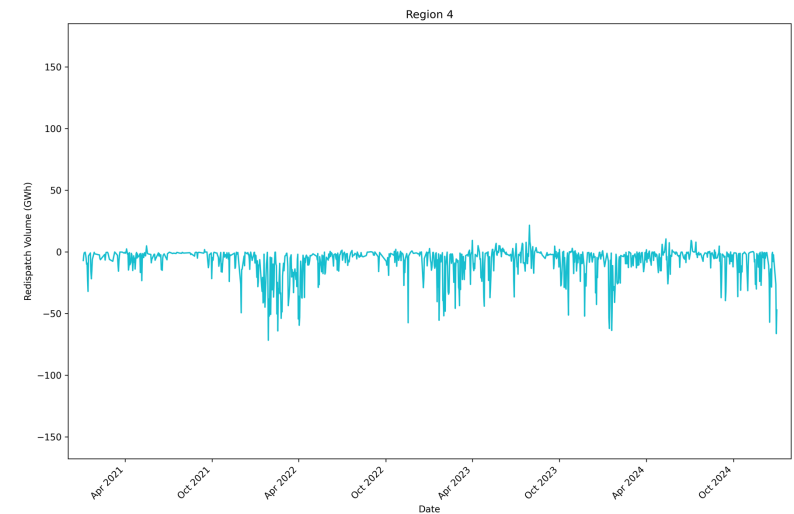
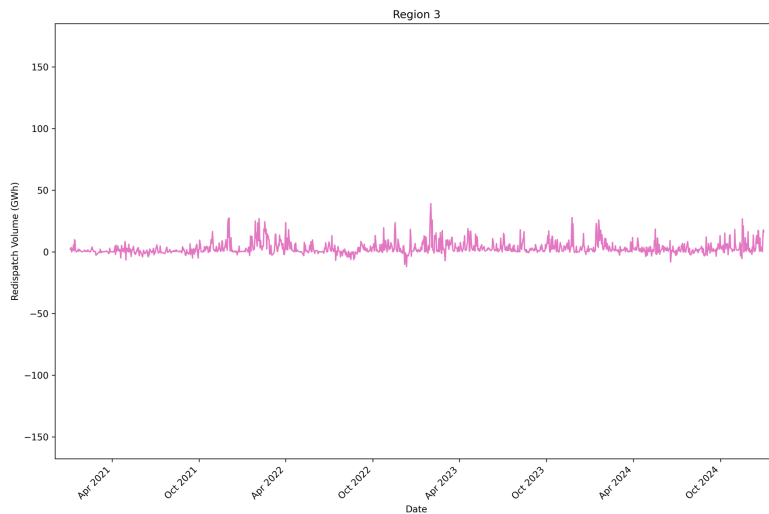
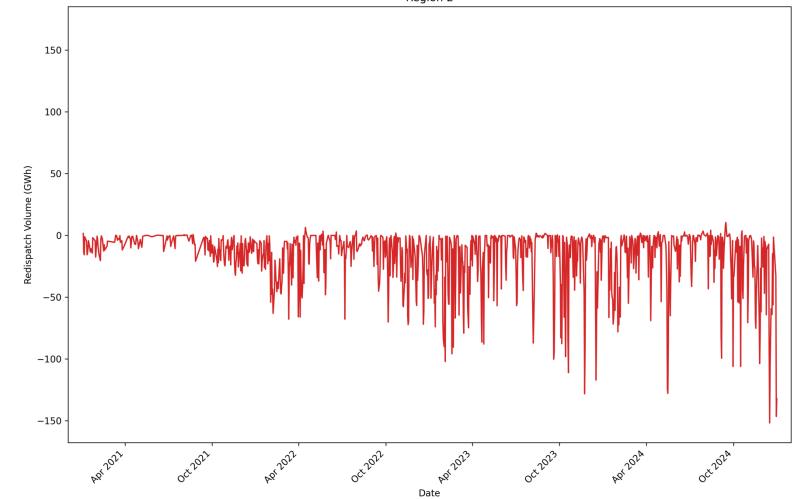
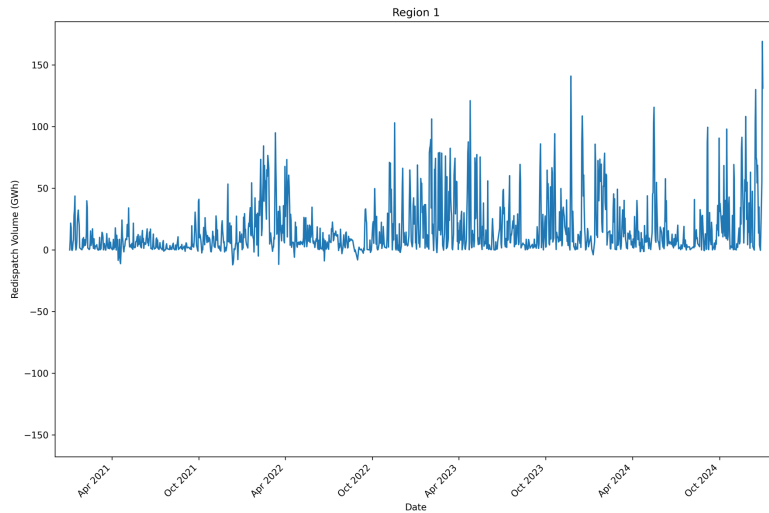
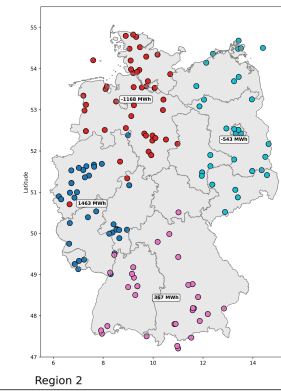
Goal: Identify regions with **consistent redispatch behavior** (direction, variance).

K-Means clustering:

- Points: redispatched units
- Features:
 - Average individual RD volume in 2021-2024
 - Geographic coordinates (lat. & long.)
- Optimal # clusters: Elbow method
- Result: **4 identified regions**



Spatiotemporal Redispatch Patterns



Probabilistic Forecasting of Redispatch Volumes

For a **given region**, consider:

- T days
- $\{y_t\}_{t \in T}$: time series of redispatch volumes
- $\mathbf{x}_t$: explanatory variables $\rightarrow$ lagged values & exogenous variables (load, generation, day-ahead prices)

Compute **day-ahead**:

- Point estimate: $\hat{y}_{t+1} = \mathbb{E}[y_{t+1} | \mathbf{x}_t]$
- Prediction interval for a quantile $\alpha \in (0, 1)$:

$$\text{PI}_{1-\alpha}(\mathbf{x}_t) = \left[Q_y \left(\frac{\alpha}{2} \mid \mathbf{x}_t \right), Q_y \left(1 - \frac{\alpha}{2} \mid \mathbf{x}_t \right) \right]$$

where the conditional quantile function is defined as $Q_y(\alpha \mid \mathbf{x}_t) = \inf \{y : F_y(y \mid \mathbf{x}_t) \geq \alpha\}$.

Models

ARIMAX

- Statistical model: autoregressive (AR), moving average (MA), and exogenous (X) components:

$$y_t = c + \beta^T \mathbf{x}_t + \sum_{j=1}^q \psi_j \epsilon_{t-j} + \epsilon_t$$

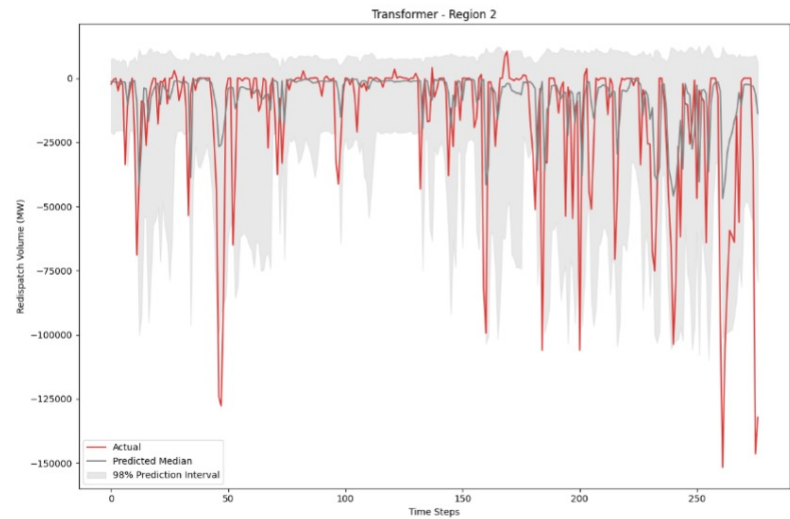
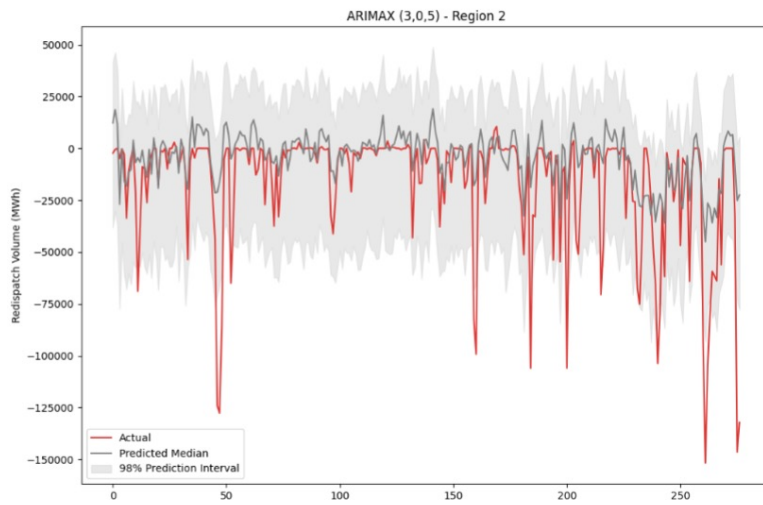
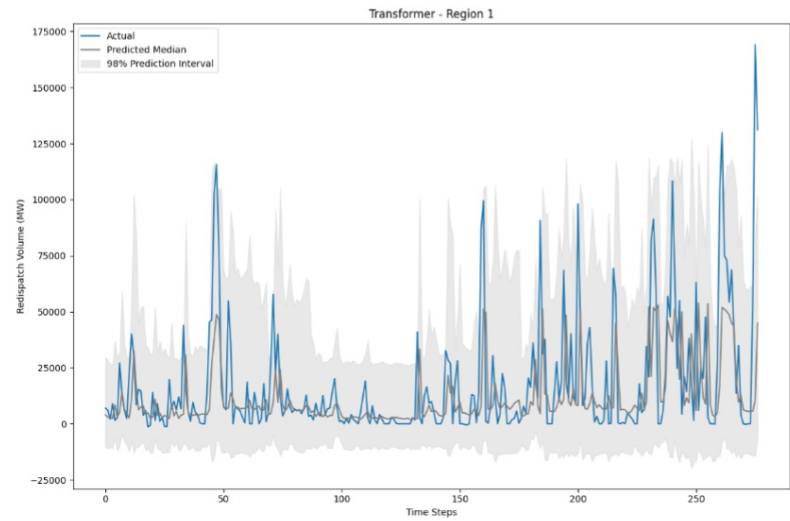
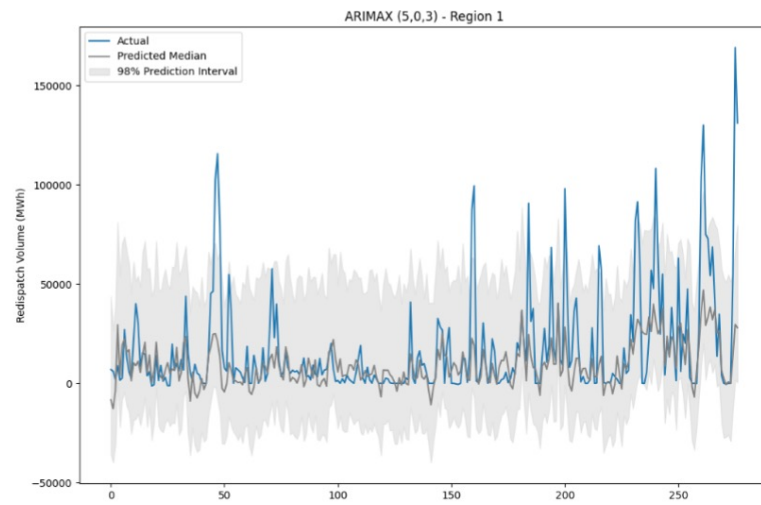
where $\mathbf{x}_t = [y_{t-1}, \dots, y_{t-p}, x_{t,1}, \dots, x_{t,d-p}]^T$ and β, ψ are model parameters.

- Steps:
 1. Fit ARIMAX model
 2. Generate point forecast
 3. 1000 x Bootstrap residuals
 4. → Empirical distribution of forecasts
- Prediction interval:** Empirical distrib. of forecasts
- Point forecast:** Median of empirical distribution

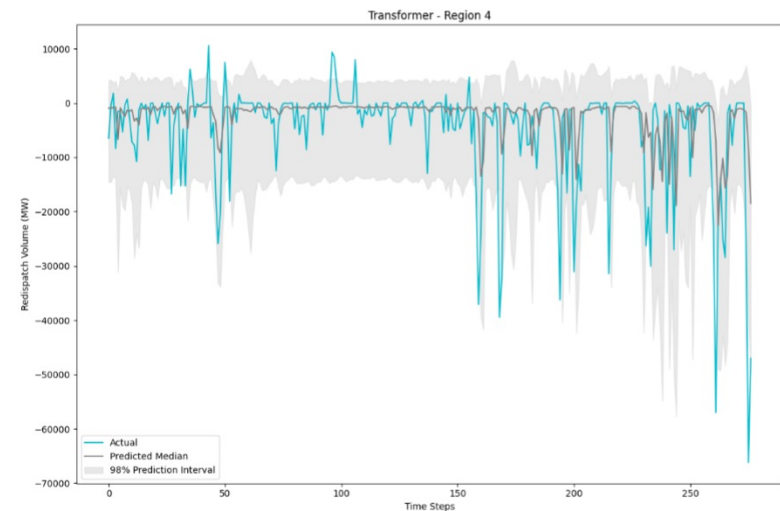
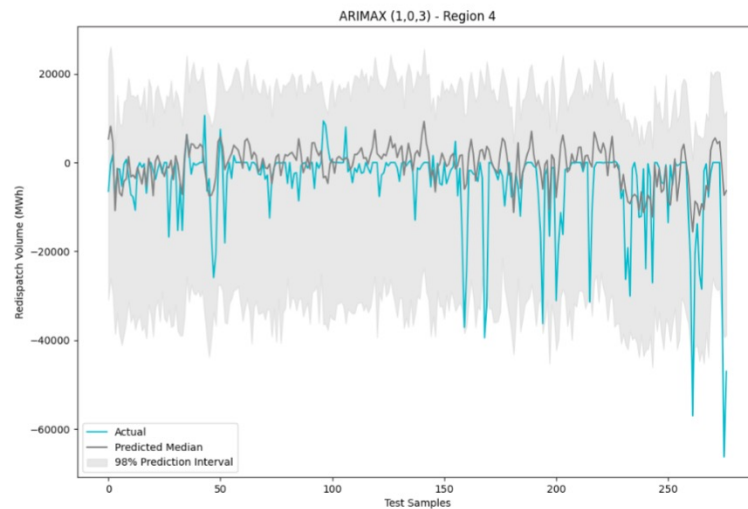
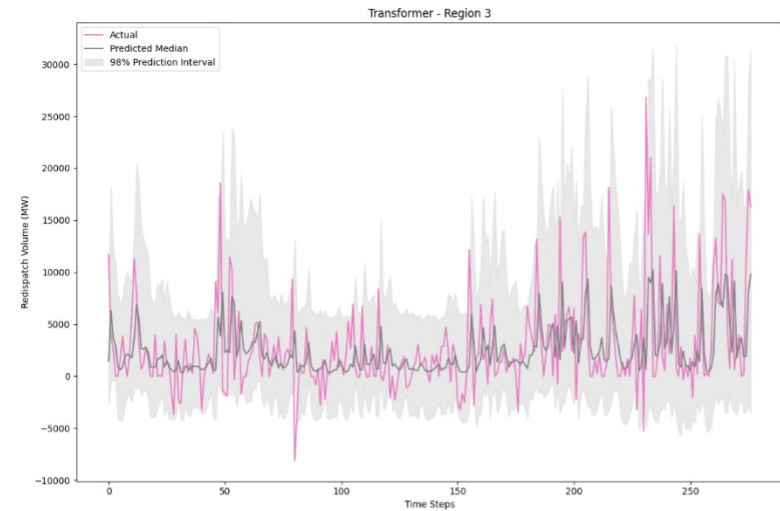
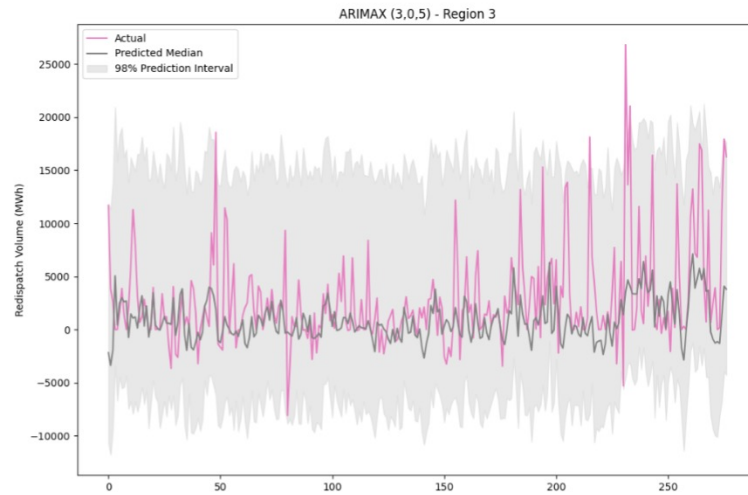
Transformer with Quantile Regression

- Deep learning architecture based on **self-attention mechanisms**.
- Input:** 15-day window $\{x_t\}_t$
- Architecture:
 1. Input sequence → linear projection + positional encoding
 2. 4 stacked Transformer encoders
 3. Select last time step representation
 4. 3 Fully connected layers
 5. Output: 3 quantiles
- Prediction interval:** $[\hat{y}^{(1)}, \hat{y}^{(q_3)}]$
- Point forecast:** $\hat{y}^{(2)}$

Forecasts: Regions 1 & 2



Forecasts: Regions 3 & 4



Results: RMSE, MAE, Empirical coverage

Reg.	Interval	RMSE		MAE		Empirical coverage	
		ARIMAX	Transformer	ARIMAX	Transformer	ARIMAX	Transformer
1	70%	22,689.81	24,496.02	12,769.64	14,039.66	0.66	0.64
	80%	22,685.59	25,097.50	12,775.22	14,626.77	0.77	0.55
	90%	22,683.34	24,195.92	12,764.60	14,139.61	0.85	0.83
	98%	22,716.22	23,784.49	12,767.49	13,406.42	0.94	0.95
2	70%	25,075.53	28,030.65	14,208.61	14,895.32	0.71	0.63
	80%	25,058.78	28,341.13	14,212.78	14,647.04	0.74	0.68
	90%	25,068.91	28,518.13	14,248.30	14,518.76	0.84	0.81
	98%	25,066.45	26,605.24	14,229.57	14,262.42	0.94	0.91
3	70%	4,615.30	4,532.55	3,050.32	2,977.13	0.60	0.53
	80%	4,615.94	4,581.42	3,042.20	2,939.82	0.77	0.65
	90%	4,625.41	4,553.16	3,062.29	2,999.08	0.89	0.77
	98%	4,606.14	4,461.61	3,042.33	2,970.75	0.99	0.91
4	70%	9,460.96	9,406.11	5,752.98	4,850.50	0.79	0.73
	80%	9,468.94	9,470.06	5,752.37	4,807.00	0.82	0.77
	90%	9,476.37	9,580.96	5,761.49	4,804.27	0.90	0.87
	98%	9,463.36	9,107.09	5,769.77	4,699.91	0.98	0.91

Summary

- Four regions with consistent redispatch behavior are identified via clustering.
- We propose a probabilistic framework for forecasting day-ahead redispatch volumes in each identified region.
- The framework integrates two complementary models:
 - ARIMAX with bootstrapped prediction intervals
 - Transformer-based neural network with quantile regression
- Model performance varies by region:
 - ARIMAX is more effective in volatile regions
 - Transformer performs best in more stable regions
- This forecasting approach provides a foundation for data-driven bidding strategies in emerging flexibility markets (Redispatch 3.0).

Thank you!